



Business Problems Solved - Telco Churn Prediction Project



1. High Customer Churn = Revenue Loss

Problem: Telecom companies lose millions due to customers switching to competitors. Acquiring new customers is 5–10× more expensive than retaining existing ones.

Solution:

- Developed a machine learning model to predict churn risk.
 - Enables early identification of high-risk customers.
 - Allows the business to proactively offer retention plans, discounts, or improved services.
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2. Lack of Visibility into Churn Drivers

Problem: Business teams often don't know why customers are leaving — whether due to pricing, poor service, or contract inflexibility.

Solution:

- Built visual analytics to explore churn patterns by:
 - Contract type
 - Tenure
 - Monthly charges
 - Services used
 - Demographics
 - Empowers data-driven marketing and customer success strategies.
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3. Slow Manual Reporting

Problem: Churn reports are often generated manually from spreadsheets or legacy systems, delaying action.

Solution:

- Built a real-time Streamlit dashboard.
 - Allows users to:
 - Predict churn instantly
 - Explore interactive charts and filters
 - Get on-demand summaries
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4. No Way to Prioritize Retention Efforts

Problem: Retention strategies often treat all customers equally, wasting budget on low-risk users.

Solution:

- Model outputs churn probability scores.
 - Created risk-based segments:
 - High risk (probability > 70%)
 - Medium risk
 - Low risk
 - Helps business focus efforts on the most at-risk customers.
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5. Lack of Model Trust and Interpretability

Problem: Stakeholders don't trust models they don't understand.

Solution:

- Used interpretable models like Logistic Regression.
 - Showed how features affect churn.
 - Provided clear evaluation metrics (precision, recall, F1-score, etc.).
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Summary Table: Business Value Delivered

Business Problem	Solution Provided
High churn = revenue loss	Predictive churn model
No visibility into churn drivers	Feature-based churn insights
Slow and manual reporting	Real-time interactive Streamlit dashboard
No targeting in retention efforts	Risk-based customer segmentation
Lack of trust in models	Transparent and interpretable ML pipeline

This project demonstrates how data science and machine learning can solve real business problems with measurable impact. While deployment was not completed due to platform limitations, all local functionality and insights are fully operational.